

Research Proposal on Agricultural Sensor Data Analysis for Crop Quality Optimization and Yield Improvement

1. Project Background and Research Objectives

With the development of agricultural modernization and digital agriculture, agricultural production is gradually shifting from traditional experience-based management to data-driven management. Crop yield and quality are influenced by multiple factors, including soil conditions, climate, irrigation, fertilization, and crop growth processes. However, many farming decisions still rely heavily on experience rather than systematic data support. As a result, it is often difficult to accurately identify the key factors affecting productivity and quality or detect production problems in a timely manner. Recent advances in agricultural sensors, the Internet of Things (IoT), and big data analytics provide new opportunities for digital agricultural management. By deploying sensors in farmland, environmental and soil conditions can be continuously monitored. Through the establishment of agricultural databases and analytical models, long-term data can be used to identify patterns influencing crop growth. Predictive models and decision-support systems can further assist farmers in making proactive management decisions that improve productivity and economic returns. This project proposes the construction of an agricultural data collection and analysis platform for a 10-acre experimental field. Real-time environmental data collected through agricultural sensors will be combined with manually collected crop growth data and final yield and quality data to establish a comprehensive agricultural big-data system. The project aims to identify the critical factors affecting crop yield and quality and develop a scalable digital agriculture management framework. **2.**

Agricultural Sensor Deployment Plan

The project will establish an agricultural IoT platform consisting of a soil monitoring system, weather monitoring system, and data transmission system. Soil monitoring equipment will include soil moisture sensors, soil temperature sensors, soil electrical conductivity sensors, and soil pH sensors. Soil moisture sensors will support irrigation management, soil temperature sensors will monitor root-zone conditions, conductivity sensors will reflect nutrient and salinity levels, and pH sensors will monitor soil acidity and alkalinity. A small agricultural weather station will be installed to continuously monitor air temperature, humidity, solar radiation, wind speed, wind direction, and rainfall. These variables provide essential information regarding external environmental conditions and serve as critical inputs for subsequent analyses. All sensors will be connected to a cloud platform through wireless networks for automated data transmission and remote monitoring. Data collection frequency is recommended at 30-minute intervals. To improve analytical reliability, the experimental field will be divided into several management zones based on location, soil characteristics, and planting conditions. Each zone will contain similar sensor configurations and standardized data collection procedures. Differential management strategies, such as varying irrigation or fertilization schedules, may be implemented in selected zones to evaluate their effectiveness through comparative analysis. **3. Data Collection and Data Management Strategy**

Data collection will be conducted at three levels: environmental data, crop growth data, and outcome data. Environmental data will be automatically collected through agricultural sensors, including soil moisture, soil temperature, soil pH, soil conductivity, air temperature, humidity, light intensity, rainfall, and wind speed. Because these variables change continuously, automated monitoring will provide millions of observations annually. Crop growth data will be collected through regular field inspections. Researchers will record plant height, leaf count, growth stages, flowering time, fruit count, fruit size, and disease or pest occurrences on a weekly basis. Drone imagery will also be collected monthly to evaluate canopy coverage and crop vigor. Outcome data will be collected at harvest. Yield indicators include yield per plant, yield per acre, and total production. Quality indicators include sugar content, fruit size, appearance scores, and product grades. Market prices and revenue information will also be recorded to enable economic performance analysis. To ensure data reliability, a data quality management framework will be established. Missing values may be estimated using neighboring observations or historical averages. Outliers will be identified and corrected through comparison with historical records. Sensors will be regularly inspected and calibrated to maintain data integrity and support accurate predictive modeling and risk analysis. **4.**

Agricultural Big Data Analytics Framework

Agricultural big data analytics represents the core component of this project. The first step involves data preprocessing, including missing-value treatment, outlier detection, normalization, and time-series alignment to establish a high-quality agricultural database. Descriptive statistical

analysis will then be conducted to understand temporal and spatial variations in environmental conditions. Trend analyses and distribution visualizations will be used to explore seasonal patterns in soil moisture, temperature, and rainfall. Subsequently, factor analysis will be performed using correlation analysis and regression modeling to evaluate relationships between environmental variables and crop performance. Examples include assessing whether soil moisture positively influences yield, whether light exposure affects product quality, and whether salinity impacts crop growth. A yield prediction model will be developed using environmental and growth-related variables as inputs and final yield as the target variable. Historical data will be used to train predictive models capable of estimating expected production before harvest, enabling more effective marketing and logistics planning. A quality prediction model will also be established to forecast quality grades using environmental conditions and growth indicators. Such predictions can help farmers adjust management practices to improve product quality. In addition, an agricultural risk warning model will be developed. Historical abnormal events will be analyzed to identify risk factors associated with yield reduction or quality deterioration. When monitored variables exceed predefined thresholds, automated warnings will be generated. Examples include irrigation alerts during prolonged drought conditions and pest management alerts during sustained hot and humid periods.

5. Agricultural Decision Support System

The ultimate objective of data analytics is to support practical decision-making rather than simply generate research findings. The proposed decision-support platform will transform complex analytical results into understandable management recommendations. Based on real-time sensor data, the system will automatically generate irrigation recommendations. When soil moisture falls below an optimal range, irrigation alerts will be issued. When significant rainfall occurs, irrigation schedules may be suspended to avoid unnecessary water use. The platform will also provide fertilization recommendations based on soil conditions and nutrient requirements. Furthermore, disease and pest risks will be estimated using environmental conditions and historical outbreak data, enabling proactive intervention before severe damage occurs. Through digital decision support, agricultural management efficiency can be improved while reducing production costs.

6. Agricultural Data Visualization Platform

To maximize the practical value of analytical results, a data visualization platform will be developed. The platform will display real-time agricultural data through dashboards, charts, and geographic visualizations. Managers will be able to monitor soil moisture, temperature, rainfall, light intensity, and crop growth conditions from computers or mobile devices. When indicators exceed predefined thresholds, graphical and textual alerts will automatically be generated. Examples include drought warnings when soil moisture remains low and pest-risk alerts during prolonged hot and humid weather. Visualization tools will make analytical results more intuitive and improve decision-making efficiency.

7. Expected Outcomes

The project is expected to produce an agricultural sensor monitoring platform, an agricultural database, yield prediction models, quality prediction models, and an agricultural decision-support system. Through long-term data collection and continuous analysis, key environmental factors affecting yield and quality will be identified and translated into actionable management strategies. From a practical perspective, the project is expected to improve resource utilization efficiency, optimize irrigation and fertilization management, reduce losses caused by diseases and pests, improve product quality, and enhance market competitiveness, ultimately increasing agricultural productivity and profitability. From a research perspective, the project will provide a real-world case study and data foundation for agricultural big data analytics, smart agriculture, and digital agriculture research, offering a replicable pathway for future agricultural digital transformation.